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XIAMEN
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DIGITAL LOGIC

Synchronous Sequential Logic Circuit

Chapter 5.4

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5.3.4 Determine the excitation and output functions and draw a logic diagram

Task: According to the binary state table and the excitation table of the selected flip-flops, find the excitation function expression of the flip-flops and the output function expression of the circuit, and simplify it; so as to construct a logic circuit that implements a given logic function with appropriate logic gates and selected flip-flops.



Flip-flop excitation table

- The excitation table of a flip-flop device illustrates the flip-flop reacts to the input (excitation) from a present state to a next state. In this table, the present state and the next state are used as independent variables, and the input is used as the dependent variables. The excitation table for the flip-flop can be derived from the flip-flop's function table.

Flip-flop excitation table

R-S Flip-flop excitation table

Q	$Q^{(n+1)}$	R	S
0	0	d	0
0	1	0	1
1	0	1	0
1	1	0	d

D Flip-flop excitation table

Q	$Q^{(n+1)}$	D
0	0	0
0	1	1
1	0	0
1	1	1

Flip-flop excitation table

J-K Flip-flop excitation table

Q	$Q^{(n+1)}$	J	K
0	0	0	d
0	1	1	d
1	0	d	1
1	1	d	0

T Flip-flop excitation table

Q	$Q^{(n+1)}$	T
0	0	0
0	1	1
1	0	1
1	1	0

According to the binary state table and the flip-flop excitation table, the simplest expression for the excitation function and the output function is generally divided into two steps:

- List the excitation and output function truth tables;
- Write the simplest expression after simplifying the Karnaugh maps.

Example: The following binary state table functions are implemented with J-K flip-flops and appropriate logic gates.

Present state y_2y_1	Next state $y_2^{(n+1)}y_1^{(n+1)}/\text{Output}$	
	$x=0$	$x=1$
00	11/0	01/0
01	00/0	00/1
10	01/0	11/0
11	00/1	10/1



A: The truth table for the excitation and output functions can be listed according to the given binary state table and the excitation table for the J-K flip-flop as shown in the table to the right.

Present state $y_2 y_1$	Next state $y_2^{(n+1)} y_1^{(n+1)}$ / Output	
	$x=0$	$x=1$
00	11/0	01/0
01	00/0	00/1
10	01/0	11/0
11	00/1	10/1

$Q Q^{(n+1)}$	J K
0 0	0 d
0 1	1 d
1 0	d 1
1 1	d 0

Excitation and Output function truth table

x	y_2	y_1	Excitation $J_2 K_2 J_1 K_1$				Output Z
0	0	0	1	d	1	d	0
0	0	1	0	d	d	1	0
0	1	0	d	1	1	d	0
0	1	1	d	1	d	1	1
1	0	0	0	d	1	d	0
1	0	1	0	d	d	1	1
1	1	0	d	0	1	d	0
1	1	1	d	0	d	1	1



Excitation and Output function truth table

 $Q Q^{(n+1)}$ $J K$

0 0

0 d

0 1

1 d

1 0

d 1

1 1

d 0

Next state

 $y_2^{(n+1)} y_1^{(n+1)}$

Excitation :

 $J_2 K_2 J_1 K_1$

Output

 Z

0

0 0

1 1

1 d 1 d

0

0

0 1

0 0

0 d d 1

0

0

1 0

0 1

d 1 1 d

0

0

1 1

0 0

d 1 d 1

1

1

0 0

0 1

0 d 1 d

0

1

0 1

0 0

0 d d 1

1

1

1 0

1 1

d 0 1 d

0

1

1 1

1 0

d 0 d 1

1

Pre

Next $y_2^{(n+1)} y_1^{(n+1)}$ /output $y_2 y_1$

x=0

x=1

00

11/0

01/0

01

00/0

00/1

10

01/0

11/0

11

00/1

10/1

Excitation and Output function truth table

x	y_2	y_1	Excitation $J_2 K_2 J_1 K_1$	Output Z
0	0	0	1 d 1 d	0
0	0	1	0 d d 1	0
0	1	0	d 1 1 d	0
0	1	1	d 1 d 1	1
1	0	0	0 d 1 d	0
1	0	1	0 d d 1	1
1	1	0	d 0 1 d	0
1	1	1	d 0 d 1	1

$x y_2$	00	01	11	10
y_1 0	1	d	d	0
1	0	d	d	0

$$J_2 = \bar{x} \cdot \bar{y}_1$$

Excitation and Output function truth table

x	y_2	y_1	Excitation $J_2 K_2 J_1 K_1$	Output Z
0	0	0	1 d 1 d	0
0	0	1	0 d d 1	0
0	1	0	d 1 1 d	0
0	1	1	d 1 d 1	1
1	0	0	0 d 1 d	0
1	0	1	0 d d 1	1
1	1	0	d 0 1 d	0
1	1	1	d 0 d 1	1

$x y_2$	00	01	11	10
y_1 0	d	1	0	d
1	d	1	0	d

$$K_2 = \bar{x}$$

Excitation and Output function truth table

x	y_2	y_1	Excitation $J_2 K_2 J_1 K_1$	Output Z
0	0	0	1 d 1 d	0
0	0	1	0 d d 1	0
0	1	0	d 1 1 d	0
0	1	1	d 1 d 1	1
1	0	0	0 d 1 d	0
1	0	1	0 d d 1	1
1	1	0	d 0 1 d	0
1	1	1	d 0 d 1	1

xy_2		00	01	11	10
y_1	0	1	1	1	1
	1	d	d	d	d

$$J_1 = 1$$

Excitation and Output function truth table

x	y_2	y_1	Excitation $J_2 K_2 J_1 K_1$	Output Z
0	0	0	1 d 1 d	0
0	0	1	0 d d 1	0
0	1	0	d 1 1 d	0
0	1	1	d 1 d 1	1
1	0	0	0 d 1 d	0
1	0	1	0 d d 1	1
1	1	0	d 0 1 d	0
1	1	1	d 0 d 1	1

$x y_2$	00	01	11	10
y_1				
0	d	d	d	d
1	1	1	1	1

$$K_1=1$$

Excitation and Output function truth table

x	y_2	y_1	Excitation $J_2 K_2 J_1 K_1$	Output Z
0	0	0	1 d 1 d	0
0	0	1	0 d d 1	0
0	1	0	d 1 1 d	0
0	1	1	d 1 d 1	1
1	0	0	0 d 1 d	0
1	0	1	0 d d 1	1
1	1	0	d 0 1 d	0
1	1	1	d 0 d 1	1

xy_1	00	01	11	10
0	0	0	0	0
1	0	1	1	1

Z

$$Z = y_2 y_1 + x y_1 = (y_2 + x) y_1$$

Based on the Karnaugh, the truth table can be used to make the excitation and output functions is shown below.

$\overline{x}y_2$	00	01	11	10
y_1 0	1	d	d	0
1	0	d	d	0

$$J_2 = \overline{x} \cdot \overline{y_1}$$

$\overline{x}y_2$	00	01	11	10
y_1 0	d	1	0	d
1	d	1	0	d

$$K_2 = \overline{x}$$

$\overline{x}y_2$	00	01	11	10
y_1 0	1	1	1	1
1	d	d	d	d

$$J_1 = 1$$

$\overline{x}y_2$	00	01	11	10
y_1 0	d	d	d	d
1	1	1	1	1

$$K_1 = 1$$

$\overline{x}y_2$	00	01	11	10
y_1 0	0	0	0	0
1	0	1	1	1

$$Z$$

$$Z = y_2 y_1 + x y_1 = (y_2 + x) y_1$$

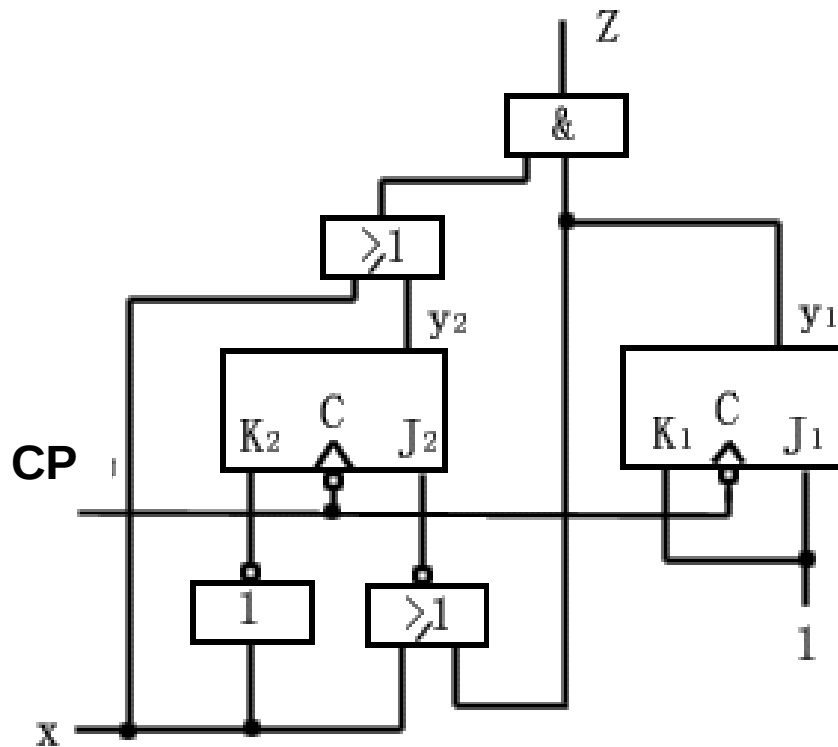


The simplified expressions of the excitation and output functions obtained by simplification are as follows:

$$J_2 = \overline{x}y_1 ; \quad J_1 = K_1 = 1$$

$$K_2 = \overline{x} ; \quad Z = y_2y_1 + xy_1 = (y_2 + x) y_1$$

The corresponding logic circuit diagram is shown below.



Practice: Using the JK flip-flops to implement the binary state table shown in the table below, try to write the excitation and output functions.

$y_1 y_2 \backslash x$	0	1
00	10,0	11,0
01	10,0	00,0
10	01,0	11,0
11	00,1	01,1

$Q Q^{(n+1)}$	J K
0 0	0 d
0 1	1 d
1 0	d 1
1 1	d 0

Practice: Use the D flip-flops to implement the binary state table, please list the excitation and output functions

Present state		Next state $y_2^{(n+1)}y_1^{(n+1)}$ / Output Z	
y_2	y_1	$x=0$	$x=1$
0	0	10/0	01/0
0	1	00/0	00/1
1	1	01/1	10/1
1	0	00/1	11/1

5.4 Synchronous sequential logic circuit design example

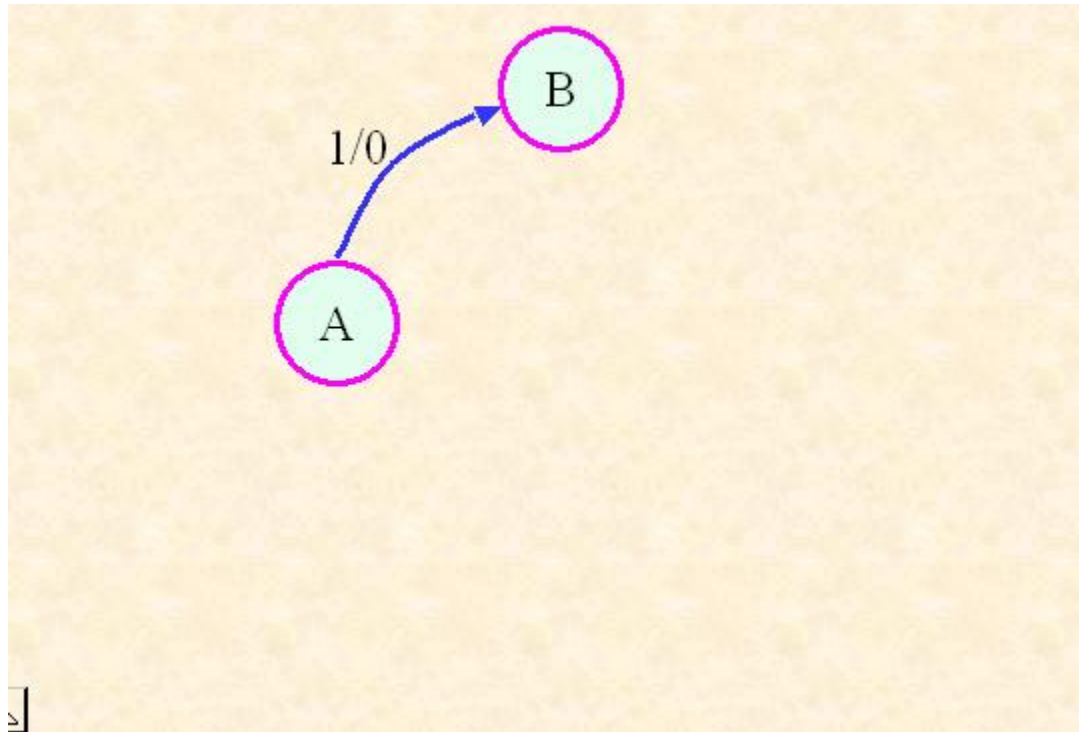
Example:

Design a “111” sequence detector, to check a sequence of binary numbers. When we have three continuous 1s, the output is 1; otherwise, the output is 0. the typical input/output are list as follows:

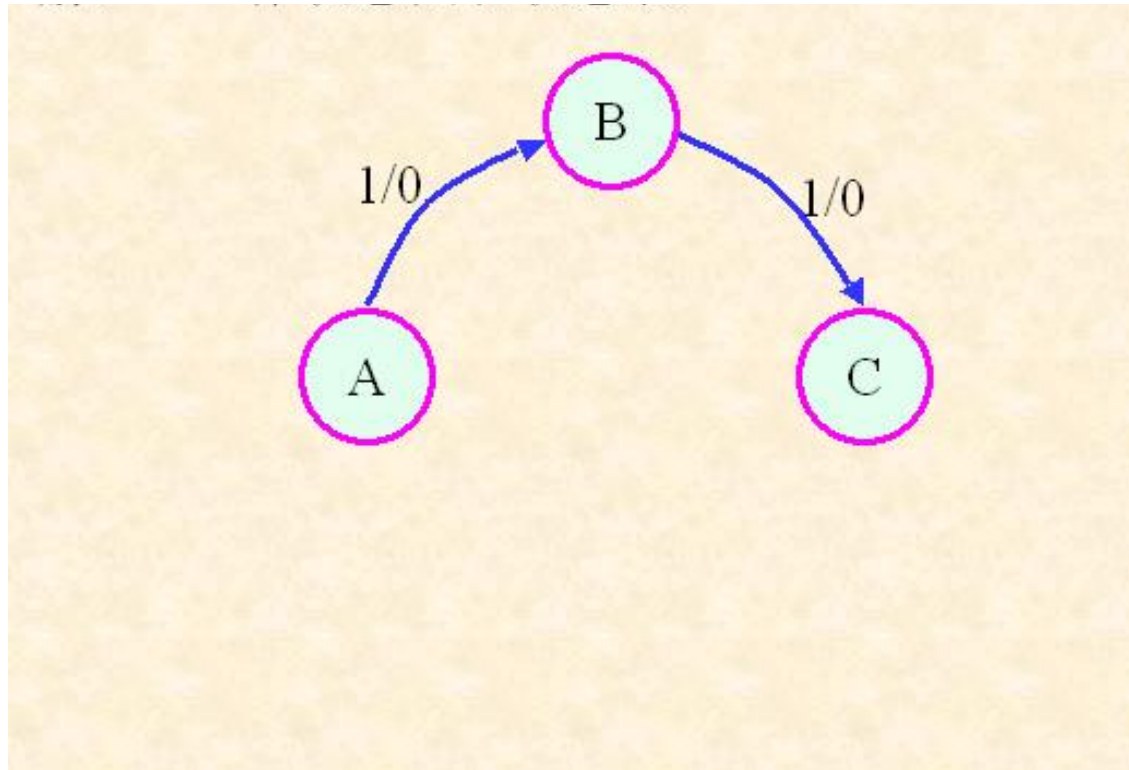
Input	X:	0	1	1	1	0	1	1	1	1	0
Output	Z:	0	0	0	1	0	0	0	1	1	0



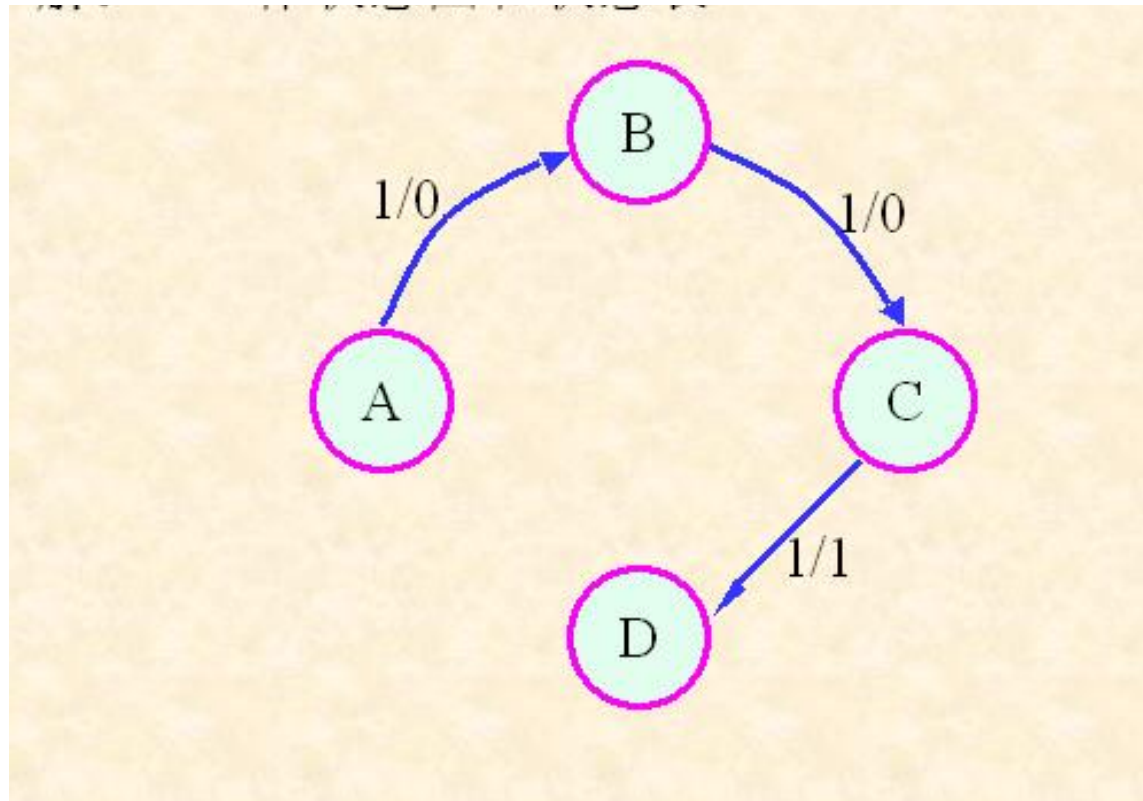
A: basic state diagram and state table



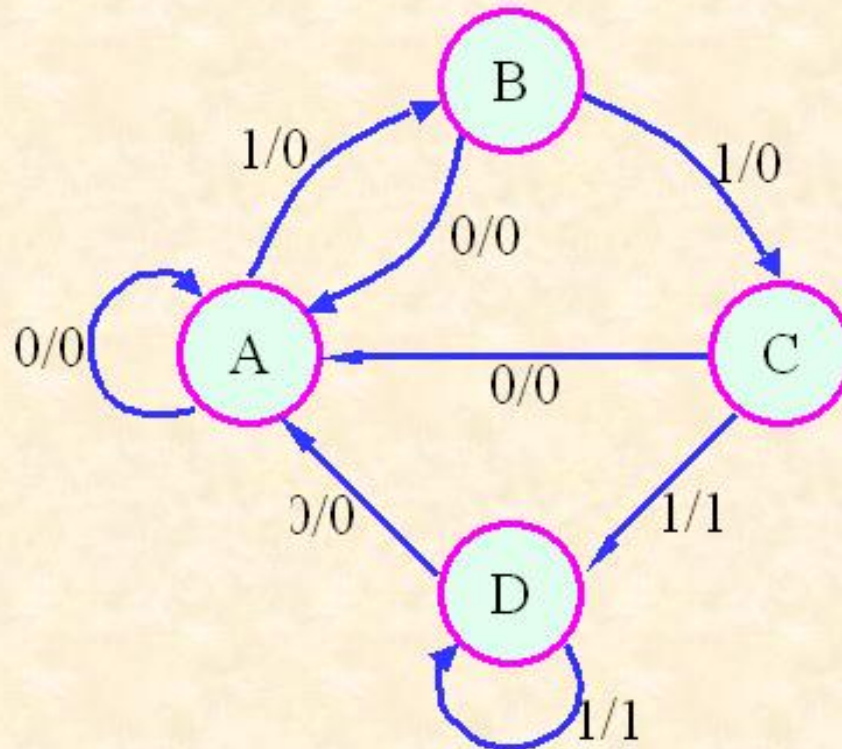
A: **basic state diagram and state table**



A: basic state diagram and state table



A: basic state diagram and state table



A: basic state diagram and state table

Present State	Next state/Output Z	
	$x=0$	$x=1$
A	A/0	B/0
B	A/0	C/0
C	A/0	D/1
D	A/0	D/1

- State reduction

The maximum equivalent classes are: {A}, {B}, {C,D}

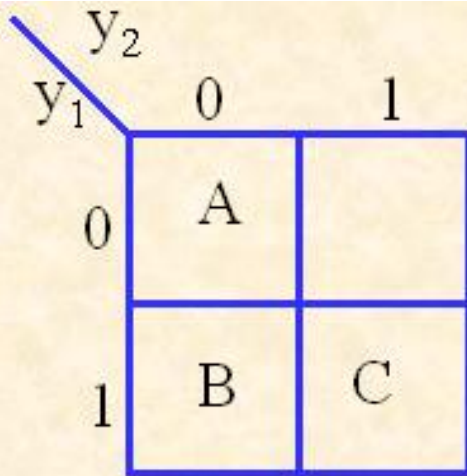
Let C to represent {C,D}, we can have the following table

Present State	Next state/Output Z		Present State	Next state/Output Z	
	$x=0$	$x=1$		$x=0$	$x=1$
A	A/0	B/0	A	A/0	B/0
B	A/0	C/0	B	A/0	C/0
C	A/0	D/1	C	A/0	C/1
D	A/0	D/1			

State Assignment:

Present State	Next state/Output Z	
	$x=0$	$x=1$
A	A/0	B/0
B	A/0	C/0
C	A/0	C/1

- A and B, B and C, A and C should be adjacent;
- A and B, A and C should be adjacent;
- A and B should be adjacent;
- A should be assigned as 0;



Present State

Next state/Output Z

$y_2 \ y_1$	$x=0$	$x=1$
0 0	00/0	01/0
0 1	00/0	11/0
1 1	00/0	11/1

Present State

Next state/Output Z

	$x=0$	$x=1$
A	A/0	B/0
B	A/0	C/0
C	A/0	C/1

Determine the excitation function and output function expressions;

Use the JK flip-flops

Q	Q ⁽ⁿ⁺¹⁾	J	K
0	0	0	d
0	1	1	d
1	0	d	1
1	1	d	0

Input	Present		Next state		Excitation				Output Z
x	y ₂	y ₁	y ₂ ⁽ⁿ⁺¹⁾	y ₁ ⁽ⁿ⁺¹⁾	J ₂	K ₂	J ₁	K ₁	Z
0	0	0	0	0	0	d	0	d	0
0	0	1	0	0	0	d	d	1	0
0	1	1	0	0	d	1	d	1	0
0	1	0	d	d	d	d	d	d	d
1	0	0	0	1	0	d	1	d	0
1	0	1	1	1	1	d	d	0	0
1	1	1	1	1	d	0	d	0	1
1	1	0	d	d	d	d	d	d	d

Input Present Excitation Output Z

x	y ₂	y ₁	J ₂	K ₂	J ₁	K ₁	Z
0	0	0	0	d	0	d	0
0	0	1	0	d	d	1	0
0	1	1	d	1	d	1	0
0	1	0	d	d	d	d	d
1	0	0	0	d	1	d	0
1	0	1	1	d	d	0	0
1	1	1	d	0	d	0	1
1	1	0	d	d	d	d	d

		xy_{200}	01	11	10
y_1					
Z	0	0	d	d	0
	1	0	0	1	0

		xy_200	01	11	10
y_1					
J_2	0	0	d	d	0
	1	0	d	d	1

	xy ₂ 00	01	11	10
y ₁				
0	d	d	d	d
1	d	1	0	d

		xy ₂ 00	01	11	10
J ₁	y ₁				
	0	0	d	d	1
	1	d	d	d	d

		xy_{200}	01	11	10
y_1					
0	K_1	d	d	d	d
1		1	1	0	0

$$J_2 = xy_1,$$

$$J_1 = x,$$

$$Z = xy_2$$

$$k_2 = \bar{x}$$

$$k_1 = \bar{x}$$

- Draw the logic circuit diagram

$$\begin{aligned} J_2 &= xy_1, & k_2 &= \bar{x} \\ J_1 &= x, & k_1 &= \bar{x} \\ Z &= xy_2 \end{aligned}$$

